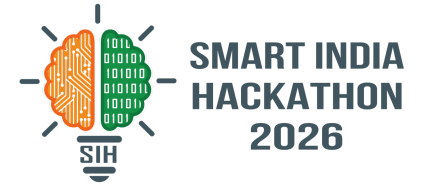


SMART INDIA HACKATHON 2026



TITLE PAGE

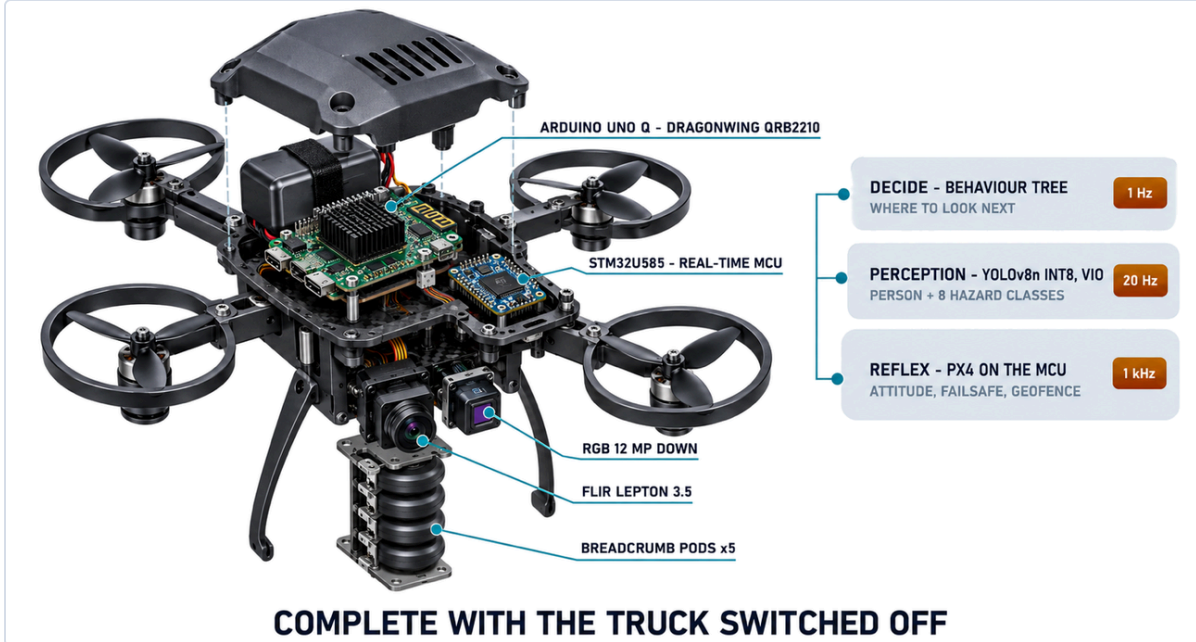
- **Problem Statement ID** – 26177
- **Problem Statement Title** – A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.
- **Theme** – Robotics and Drones
- **PS Category** – Hardware
- **Team ID** –
- **Team Name** – Drone Syndicate



PROPOSED SOLUTION

At Wayanad, 206 people were never found. Not because nobody looked. It was because 1,300 responders could not cover 15 km² before the window closed.

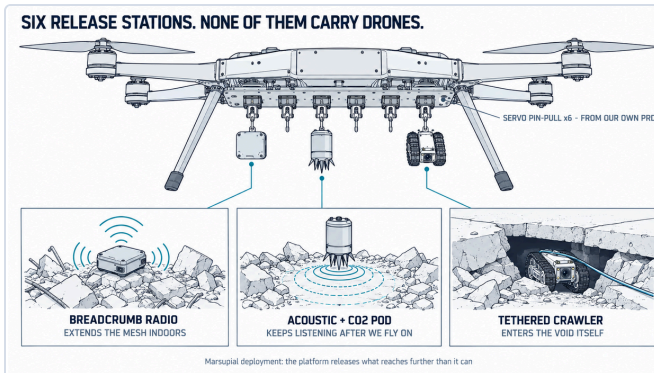
① ONE AIRCRAFT · COMPLETE WITH THE TRUCK SWITCHED OFF



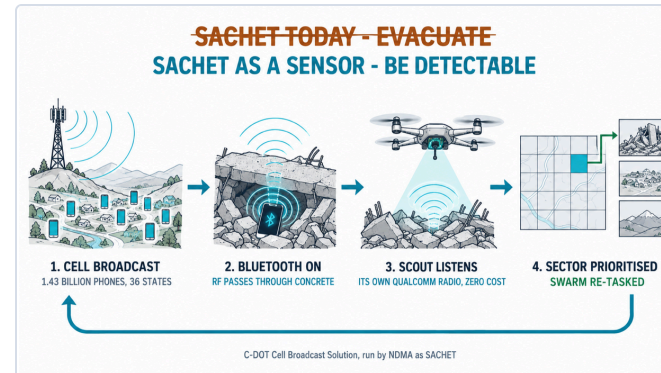
② SIX OF THEM · 15 KM² SWEEPED IN 5.8 HOURS



③ ONE AIRCRAFT, THREE PAYLOADS



④ THE IDEA NOBODY HAS USED



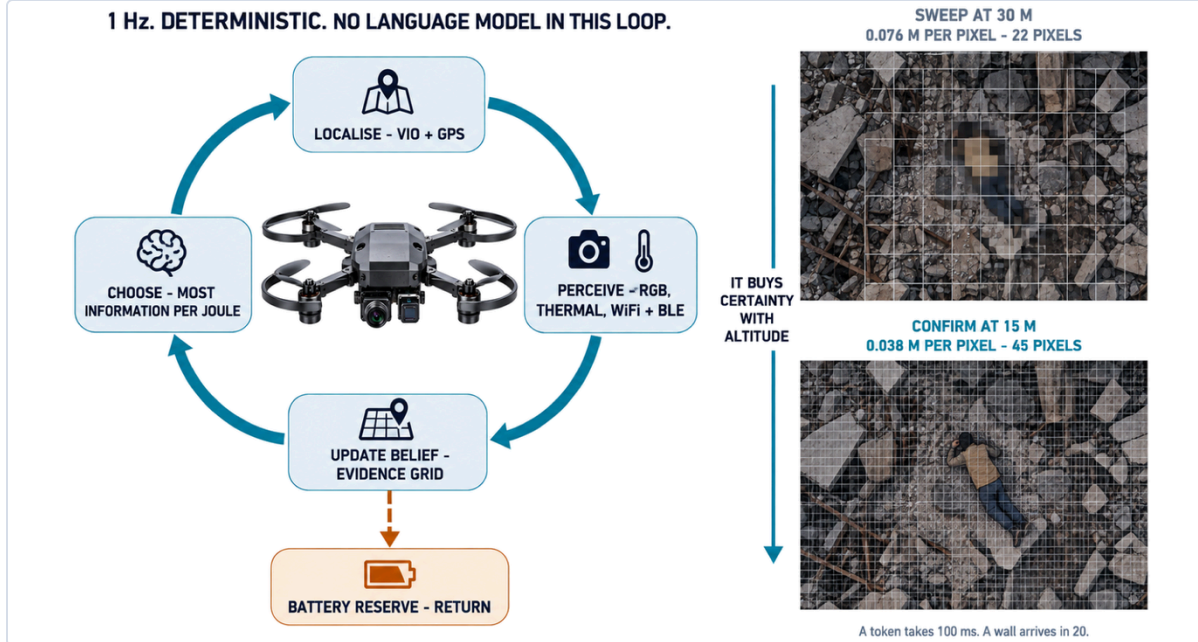
◆ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation:** One scout is an **Arduino UNO Q** (Qualcomm Dragonwing QRB2210 + real-time STM32U585) carrying RGB, thermal and its own Wi-Fi/BLE radio. PX4 flies on the MCU at 1 kHz; detection, odometry and the decision tree run on Linux. **It is not only a swarm.** A single aircraft also carries swappable payloads: breadcrumb radios, acoustic and CO₂ pods, a tethered crawler. The truck is an accelerator, never a dependency.
- How it addresses the problem:** Wayanad 2024: 1,300 responders, 6 zones, **5+ days**, 206 never found. One scout sweeps 0.43 km²/h alone; six sweep the full 15 km² in **5.8 hours**, for **+64 survivors per 100 trapped**.
- Innovation and uniqueness:** **SACHET cell broadcast, inverted.** NDMA already reaches 1.43 bn handsets to say *evacuate*. One message saying *turn Bluetooth on* makes every phone a beacon, and the scout's own Qualcomm radio is already the receiver. **Zero added hardware.**

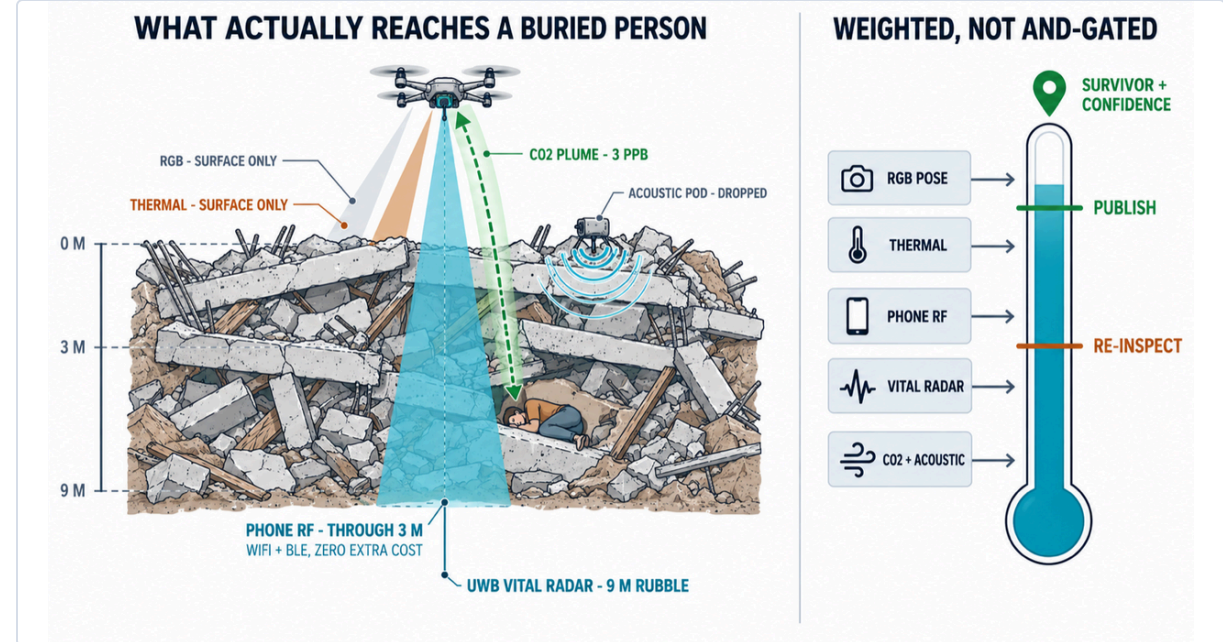
TECHNICAL APPROACH

The detector is ours. The decision layer is ours. Both are measured on Qualcomm silicon, and the aircraft has flown the profile.

① HOW ONE AIRCRAFT DECIDES · 1 HZ, DETERMINISTIC



② WHAT ACTUALLY REACHES A BURIED PERSON



RUNS IN THE JUDGE'S BROWSER



Live demo
hf.co/spaces/anshumanatrey/
kestrel-survivor-detection

◆ Technical Approach

SILICON **Dragonwing QRB2210** **QCS6490** **IQ-8275** **STM32U585** AI **YOLOv8n · SARD fine-tuned** **MobileCLIP2-S0** **Qualcomm AI Hub** **QNN** FLIGHT **PX4** **ROS 2** **VINS-Fusion VIO**

RADIO **Wi-Fi 5** **BT 5.1** **LoRa SX1262** **breadcrumb relays**

- **Methodology, four algorithms not a wrapper:** (1) two-pass descent, SAHI-style, with min-window-cover crop planning, union-find cross-pass merging and weighted box fusion · (2) correlated Bayesian fusion solving $\Sigma w = 1$, discounting sensors that fail together and reducing to Chair-Varshney when they do not · (3) prize-collecting **Held-Karp** route DP, exact, battery reserve inside the DP · (4) 0/1 knapsack for the LoRa frame. **9/9 self-tests against brute force.**
- **Trained, not borrowed:** YOLOv8n fine-tuned on **SARD**, the published aerial SAR benchmark (1,980 images, 6,525 person instances), 183 epochs on a T4. Held-out test: **mAP@50 0.952**, precision 0.969, recall 0.900. Literature puts YOLOv4 at 97.15% mAP@0.4 on the same data; ours is 95.2% at the stricter @0.5.
- **Measured and flown, not estimated:** **Qualcomm AI Hub, physical devices.** Our detector INT8 @640 on Arduino Ventuno Q: **1.72 ms / 580 FPS**, MobileCLIP2-S0 beside it **0.87 ms**, **two networks, 2.59 ms per frame, 386 Hz** against a sweep needing 0.22 Hz, **480 of 480 layers on the Hexagon NPU**. PX4 SITL flew the profile: **30.0 m, 3.8 m/s, a 220 × 90 m lane grid**. Public log, slide 6.

We built four architectures, scored them against our own criteria, and shipped the one that survives losing pieces.

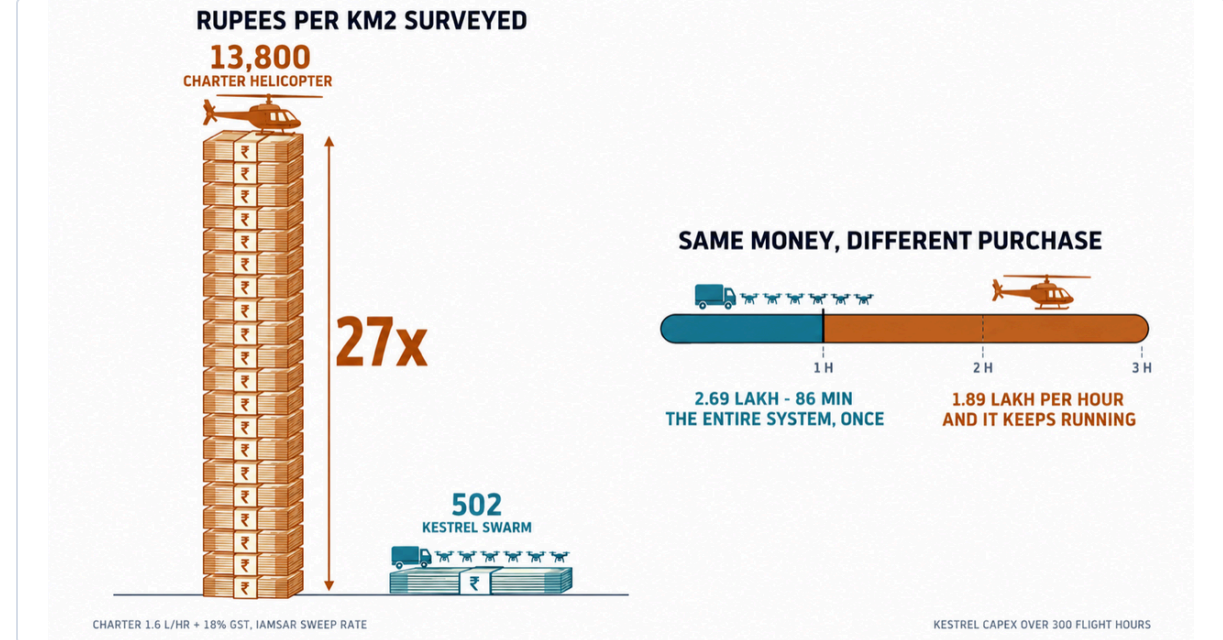
① THE ALTERNATIVES WE TESTED · AND WHY OURS WON

FOUR WAYS TO BUILD THIS. WE TESTED OURS.

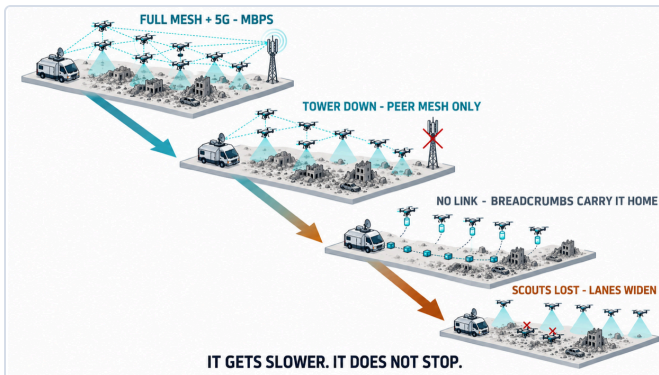
	A. FLYING MOTHERSHIP	B. TRUCK IS THE BASE	C. PEER SWARM	D. SWARM IS THE COMPUTER
NO SINGLE POINT OF FAILURE	✗	✓	✓	✓
COMPUTE FULLY USED	✗	✓	✗	✓
WORKS INSIDE BUILDINGS	✗	✓	✓	✓
EVERY NODE QUALCOMM	✓	✓	✓	✓
	OUR FIRST DESIGN			

Evidence: Zhejiang Science Robotics 2022; CERBERUS DARPA SubT; ACHORD; market-based replanning 2026

② WHAT IT COSTS AGAINST THE ALTERNATIVE



③ WHEN PIECES FAIL



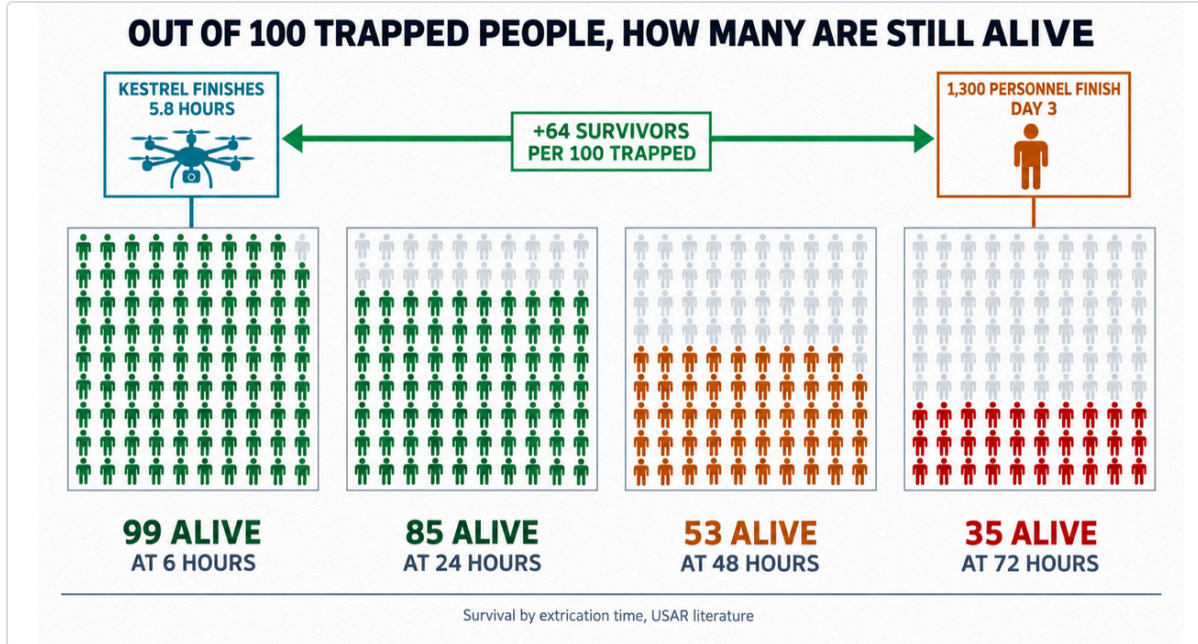
◆ Feasibility and Viability

- **Feasibility:** ₹2.68 lakh total, Indian-vendor sourced (Robu / Robocraze / Thundercomm). That is **86 minutes** of charter helicopter. Airframe, flight stack and radios are all commodity; the detector is already profiled on real Qualcomm hardware.
- **Potential challenges and risks:** Thermal **inverts** on sun-heated rubble in daylight · VIO drifts ~0.5% of path, which is 21 m over a full sortie · FINDER-class radar **reported three breath signals at Wayanad where nothing was found** · a 500 mm airframe cannot enter a collapsed doorway.
- **Strategies for overcoming them:** Independence weighting means **no single stream ever decides**. The radar that lied is one weighted vote · GPS-denied excursions are bounded to 1–3 min by breadcrumb anchors and loop closure at the truck, giving **1.2–3.6 m**, inside the re-inspect radius · interiors are reached by **dropped breadcrumb pods**, not by flying the sweep frame indoors · and the degradation ladder is designed so the system gets **slower, never offline**.

IMPACT AND BENEFITS

The 72-hour window is the entire problem. We move first contact from day three to hour six.

① OUT OF 100 TRAPPED PEOPLE, HOW MANY ARE STILL ALIVE



② WHAT THE COMMANDER SEES · ON-DEVICE, NO NETWORK



③ WHAT IT IS ALL FOR



◆ Impact and Benefits

- Impact on the target audience:** NDRF, SDRF and the Army search at roughly 0.1 km²/h per team. Kestrel sweeps **2.6 km²/h**. On the Wayanad footprint that is **5.8 hours instead of five days: 99 alive per 100** at six hours against **35 at seventy-two**.
- Social:** Responders stop walking unstable debris and live power lines to find out whether anyone is there. The aircraft takes that risk, and it takes it at night, in monsoon, without fatigue.
- Economic:** ₹502 per km² against ₹13,800 by charter helicopter, **27×**. Equipping all **738 districts** costs ≈ **₹19.86 Cr**, which is **0.12%** of the national disaster preparedness line. It is already funded; it is a procurement decision, not a budget request.
- Environmental:** Battery-electric, no fuel burn, and the scout's brain is a ₹5,190 board rather than a ₹50,000 one.

Every number in this deck traces to a source. Every claim about our own silicon is a job ID you can open.

PEER-REVIEWED & FIELD EVIDENCE

- ▶ Team **CERBERUS**, winner, DARPA Subterranean Challenge 2021: base-station compute + breadcrumbed radio nodes
- ▶ Zhejiang Univ., *Swarm of micro flying robots in the wild*, **Science Robotics 2022**: decentralised, no central aircraft
- ▶ Akyon et al., **SAHI**, [arXiv 2202.06934](#): slicing raises AP +6.8% on VisDrone with no retraining
- ▶ Solovyev et al., **Weighted Boxes Fusion**, [arXiv 1910.13302](#)
- ▶ **Chair & Varshney**, IEEE TAES 1986: optimal LLR decision fusion; we generalise it to correlated streams
- ▶ *Sensors 18(3) 852*: CO₂ + thermal + microphone fusion for trapped-victim detection
- ▶ NASA JPL / DHS **FINDER**: 9 m rubble penetration; four men found alive, Nepal 2015
- ▶ **ACHORD** / CARA (JPL CoSTAR): communication-aware coordination with droppable radios
- ▶ Held & Karp, J. *SIAM* 1962: exact DP for the sequencing problem we solve per sortie
- ▶ UAV-VLRR [arXiv 2503.02465](#) · **AVERY** [arXiv 2511.18151](#): VLM split computing for disaster response



Live demo · runs in your browser
Two-pass descent, correlated fusion, Held-Karp routing and the knapsack link budget, executing on *your* device. No server, no login.
huggingface.co/spaces/anshumanatrey/kestrel-survivor-detection

INDIAN OPERATIONAL & POLICY RECORD

- ▶ **Wayanad landslide**, 30 Jul 2024: 1,300 personnel, 40 teams, 6 zones, 5+ days, 357 dead, **206 missing**, ~15 km²
- ▶ **C-DOT SACHET** / NDMA cell broadcast: 1.43 bn citizens, 36 states, 14.5 M tower cells
- ▶ **15th Finance Commission** 2021–26: NDRF/SDRF preparedness allocation
- ▶ **DGCA** UAS Rules 2021 · Drone Rules amendment: micro/small category, RPC training
- ▶ [keralarescue.in](#), Kerala 2018: crowd-sourced rescue requests, IEEE Kerala + Kerala IT Mission
- ▶ **Nepal–Tibet glacial flood**, Aug–Sep 2026 · **Assam floods**, Jul–Aug 2026: the live monsoon record
- ▶ **NDMA** / NDRF deployment doctrine: 40 teams across 6 zones is the Wayanad baseline we measure against
- ▶ **IAMSAR** sweep-width methodology: the ground-team rate our 0.1 km²/h figure derives from
- ▶ **ISRO Bhuvan** · Copernicus **Sentinel-1 SAR**: free pre-flight damage priors

QUALCOMM PLATFORM & OUR OWN MEASUREMENTS

- ▶ **Qualcomm AI Hub**: YOLOv8n INT8 @640, QNN DLC, `--quantize_io`, profiled on physical devices
- ▶ Arduino **Ventuno Q** (IQ-8275, Hexagon v75), our SARD detector INT8: **1.72 ms · 580 FPS** · **247/247 NPU** · MobileCLIP2-S0: **0.87 ms · 233/233 NPU**
- ▶ Dragonwing **RB3 Gen 2** (QCS6490, Hexagon v68), our SARD detector INT8: **10.77 ms · 93 FPS** · **247/247 NPU**; fp16 and MobileCLIP will not compose on v68
- ▶ Arduino **UNO Q**: QRB2210 + STM32U585, Wi-Fi 5 + BT 5.1, ₹5,190 (4 GB / 32 GB)
- ▶ **PX4 / ROS 2 / VINS-Fusion**: open flight and odometry stack
- ▶ **PX4 Flight Review**: [public flight log](#), PX4 v1.15.4 SITL, 30.0 m · 3.8 m/s · 220 × 90 m survey grid
- ▶ *Third-party*: Foundries.io measured YOLOv5 Pico at **~17 FPS** on a physical UNO Q
- ▶ **SARD** aerial search-and-rescue dataset, 1,980 images, 6,525 person instances: **fine-tuned**, **mAP@50 0.952** on the held-out test split, 183 epochs, Tesla T4
- ▶ **ONNX Runtime Web**: the demo executes in the judge's browser, no server and no upload
- ▶ *Honest limit*: **QRB2210 is not offered on AI Hub**, so the scout figure stays third-party



Engineering record
Sourced data registry, BOM, every prompt, and the algorithm module with its brute-force self-tests.
github.com/AnshumanAtrey/sih2k26-the-drone-syndicate

OUR OWN MEASUREMENTS · QUALCOMM AI HUB, PHYSICAL DEVICES, JOB IDS ON REQUEST

DEVICE	SILICON	PRECISION	LATENCY	THROUGHPUT	COMPUTE UNITS	PEAK MEMORY
Arduino Ventuno Q · our SARD detector	Dragonwing IQ-8275 · Hexagon v75	INT8	1.72 ms	580.7 FPS	247 / 247 NPU	6.3 MB
Arduino Ventuno Q · MobileCLIP2-S0 visual	Dragonwing IQ-8275 · Hexagon v75	INT8	0.87 ms	1146.8 FPS	233 / 233 NPU	4.8 MB
Arduino Ventuno Q · both networks, per frame	Dragonwing IQ-8275 · Hexagon v75	INT8	2.59 ms	386 Hz	480 / 480 NPU	11.1 MB
Dragonwing RB3 Gen 2 · our SARD detector	QCS6490 · Hexagon v68	INT8	10.77 ms	92.8 FPS	247 / 247 NPU	8.4 MB
Arduino Ventuno Q · stock COCO, for comparison	Dragonwing IQ-8275 · Hexagon v75	INT8	1.85 ms	539.7 FPS	247 / 247 NPU	7.5 MB
<i>Dragonwing RB3 Gen 2 · fp16, and MobileCLIP</i>	<i>QCS6490 · Hexagon v68</i>	–	<i>graph will not compose on Hexagon v68, twice: reported as negative results, not omitted</i>			